

## Topic 5

### Kinetic theory and gases

- 5.1 Use a simple kinetic theory model to describe movement of particles in the three states of matter
- 5.2 Explain the pressure of a gas in terms of the motion of its particles
- 5.3 Describe the effect of changing the temperature of a gas on the speed of its particles
- 5.4 Describe the term absolute zero,  $-273^{\circ}\text{C}$ , in terms of the lack of movement of particles
- 5.5 Convert between the Kelvin and Celsius scales
- 5.6 Recall that the average kinetic energy of the particles in a gas is directly proportional to the Kelvin temperature of the gas
- 5.7 *Investigate the temperature and volume relationship for a gas*

- 5.8 Use the relationship:

$$V_1 = \frac{V_2 T_1}{T_2}$$

to calculate volume for gases of fixed mass at constant pressure  
(rearranging not required)

- 5.9 *Investigate the volume and pressure relationship for a gas*

- 5.10 Use the relationship:

$$V_1 P_1 = V_2 P_2$$

to calculate volume or pressure for gases of fixed mass at  
constant temperature

- 5.11 **Use the equation:**

**initial pressure (pascal, Pa) × initial volume (metre<sup>3</sup>, m<sup>3</sup>)  
/ initial temperature (kelvin, K) = final pressure (pascal,  
Pa) × final volume (metre<sup>3</sup>, m<sup>3</sup>) / final temperature  
(kelvin, K)**

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

- 5.12 Apply an understanding of the equation in 5.11 to the use of  
bottled gases in medicine, including the need for a pressure  
above atmospheric and the calculation of the volume of gas  
released at atmospheric pressure

# Unit PCA: Physics controlled assessment

## Overview

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### Content and How Science Works overview

The controlled assessment is designed to enable students to engage with the scientific process through setting a hypothesis relevant to a given set of variables and then planning an investigation, observing, recording and presenting outcomes and conclusions.

The student task will consist of three parts:

#### **Part A – Planning**

#### **Part B – Observations**

#### **Part C – Conclusions**

The tasks, provided by Edexcel, will relate to the following units in this specification:

P2 – Physics for your future

P3 – Applications of physics

**Students must NOT submit a controlled assessment task for P1 for this qualification.**

The quality of written communication will be important in all reports produced as how students present, order and explain their work links directly to how well it is understood by the reader.

### Assessment overview

- This unit is internally assessed under controlled conditions.
- There will be two tasks available each year – one task from P2 and one task from P3.
- Each task has a shelf life of one year.
- The tasks will be available to teachers one year in advance.
- Each task has three parts – Part A: Planning, Part B: Observations and Part C: Conclusions.
- The total number of marks available for the three parts is 50.
- Students must attempt all three parts of a task.
- If they attempt both tasks, then the best marks from Part A, B and C should be submitted for the unit.

### Detailed unit content

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#### Delivery of the controlled assessment

##### Skills

Students should demonstrate the ability to carry out the following skills when completing a task:

- a develop a hypothesis and plan practical ways to test it including risk assessment
- b manage risks when carrying out practical work
- c collect, process, analyse and interpret primary and secondary evidence including the use of appropriate technology to draw evidence-based conclusions
- d review methodology to assess fitness for purpose, and review the hypothesis in light of outcomes.

##### Parts of the controlled assessment tasks

###### Part A – Planning (20 marks)

Includes choosing equipment, hypothesis, controls needed for the task, evidence/observations and range, identification and management of risk.

###### Part B – Observations (6 marks)

Includes primary and secondary evidence collection and recording.

###### Part C – Conclusions (24 marks)

Includes processing and presentation of evidence, quality of evidence, conclusions based on evidence, evaluation of method, evaluation of conclusion.

##### Student support

Where students produce a plan that is unworkable or dangerous, it is permitted for teachers to provide students with a plan, provided it is clear that students will not receive Part A marks for this plan.

## Unit PCA: Physics controlled assessment

### Levels of control

Internal assessment under controlled conditions has levels of control for task setting, task taking and task marking. These must be adhered to when students are completing their controlled assessment tasks.

#### Summary of levels of control

| Area                  | Level of control |
|-----------------------|------------------|
| Part A – Planning     | Limited          |
| Part B – Observations | Limited          |
| Part C – Conclusions  | High             |

### Task setting

#### High level of control

A high level of control means that tasks will be set by Edexcel and centres will choose from a list of tasks, from the other units in this qualification.

The tasks will change every year, in accordance with the Ofqual regulations for GCSE Science. Teachers must take care when using these tasks to ensure that students are completing the correct task for a particular year. The front sheet of each task will show the dates for which it is valid.

#### When will the tasks be available?

They will be available on the Edexcel website for teachers to download a year ahead of their first assessment opportunity. Teachers can view all the task sheets available before deciding which task the students will complete.

#### When should the tasks be made available to students?

The task sheets for this controlled assessment are confidential and must not be shown to students before they start the tasks. Task sheets should not be shown to students until the start of the task planning stage of the controlled assessment.

#### Do all my students have to do the same task?

It is acceptable for all the students in a class to complete the same task. However, the same task does not have to be chosen for all students and they can work on a mixture of different tasks from P2 and P3.

### Task taking

#### **a Research and data collection – limited level of control**

Research and data collection, including practical work, will be carried out under limited control. This means that students may work collaboratively when collecting data from practical activities.

Students may carry out any secondary research whilst not being directly supervised by a teacher, for example in a library or at home. The secondary research can include extracts from books and websites.

#### **b Analysis, conclusions and evaluation of findings – high level of control**

The analysis, conclusions and evaluation will be produced by students under high levels of control. This means that students must carry out this part of the write-up individually, under the supervision of a teacher.

The production of the final report will usually take place over several lessons, so the students' materials must be collected in at the end of the lesson and handed back at the beginning of the next one. Students' final reports must be produced individually.

### Communication with students during the controlled assessment

Feedback can be given to students during the controlled assessment, but this must be general rather than specific feedback. Teachers may give students general feedback on:

- the equipment chosen
- the controls for the task
- data to be collected or observations to be made
- risks involved with the task
- techniques for processing data/observations
- skills involved in the conclusions and evaluation.

Students should receive a copy of the assessment criteria so that they are aware of what they need to do to access the full range of marks.

### Suggested timings of each area

The suggested timings for each part of the controlled assessment task are as follows:

|                       |        |
|-----------------------|--------|
| Part A – Planning     | 1 hour |
| Part B – Observations | 1 hour |
| Part C – Conclusions  | 1 hour |
| Total of 3 hours      |        |

For this controlled assessment unit, it is expected that students should be given approximately 6 hours of time specifically on preparing for tasks. By using the practicals noted in the theory units, this can be achieved as part of your normal teaching.

## Unit PCA: Physics controlled assessment

### Task marking

#### Task marking – medium level of control

A medium level of control means that the marking of the tasks will be carried out by teachers and moderated by Edexcel.

### Marking procedure

Teachers should use the assessment criteria to mark the tasks and use the *Controlled Assessment Record Sheet* (Appendix 5) to record the marks. Edexcel will give additional guidance of the application of the generic marking criteria in support documentation.

It is good practice for teachers to annotate students' work to show how the marks have been allocated for each section.

### Submitting marks

Students must attempt all three parts of any task they do.

Final marks for each section of the students' work should be recorded on the *Controlled Assessment Record Sheet* in Appendix 5.

They don't need to submit all marks from a task but can submit the best marks from any of the tasks they have attempted.

If a mark is submitted from a task, the student response to all three parts must be marked and retained by the centre for moderation.

Each CAT may be submitted for moderation in either May (for a June session) or October (for a November session).

### Health and safety

Students must observe safe practice when they are carrying out practical work. It is the responsibility of centres to carry out risk assessments for all practical work that they undertake with their students.

In this internal assessment teachers will have limited control when students are collecting their data, but it should be carried out under full supervision for health and safety reasons. The limited control means that students can work collaboratively to collect their data.

## Assessment criteria

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### Part A - Planning

| Element   | Marks |           | Criteria                                                                                                                |
|-----------|-------|-----------|-------------------------------------------------------------------------------------------------------------------------|
| Equipment | 2     | 0 marks   | Gives no relevant detail                                                                                                |
|           |       | 1–2 marks | a) Explains most relevant resources/equipment<br>b) Explains reasons for choices and choices are fully relevant to plan |



## Unit PCA: Physics controlled assessment

| Element                                                                                                                                                                                                                                 | Marks |           | Criteria                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Controls</b><br>(If variables are to be controlled, criteria a1 and b1 will be used. If there are no variables to control, criteria a2 and b2 will be used. The specific criteria needed will be in the controlled assessment task.) | 6     | 0 marks   | Gives no relevant controls                                                                                                                                                                                                                                                                         |
|                                                                                                                                                                                                                                         |       | 1–2 marks | a1) Identifies one appropriate variable to control<br>b1) Describes how this variable can be controlled<br>OR<br>a2) Identifies one appropriate way to control the task<br>b2) Describes this way of controlling the task                                                                          |
|                                                                                                                                                                                                                                         |       | 3–4 marks | a1) Identifies some relevant variables to control<br>b1) Gives an appropriate description of how to control these variables<br>OR<br>a2) Identifies some relevant ways to control the task to produce meaningful results<br>b2) Describes how these ways control the task                          |
|                                                                                                                                                                                                                                         |       | 5–6 marks | a1) Identifies a range of variables appropriate to control<br>b1) Gives an appropriate explanation of how to control these variables<br>OR<br>a2) Provides a comprehensive list of relevant ways to control the task to produce meaningful results<br>b2) Explains how these ways control the task |

## Unit PCA: Physics controlled assessment

| Element             | Marks     |           | Criteria                                                                                                                                                                                                                        |
|---------------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hypothesis</b>   | 4         | 0 marks   | Provides no relevant hypothesis                                                                                                                                                                                                 |
|                     |           | 1–2 marks | a) Provides hypothesis that is appropriate for the scope of the task<br>b) Partially justifies the hypothesis                                                                                                                   |
|                     |           | 3–4 marks | a) Provides hypothesis that is appropriate for the scope of the task based on relevant scientific ideas<br>b) Justifies the hypothesis fully using relevant scientific ideas                                                    |
| <b>Risks</b>        | 4         | 0 marks   | No relevant detail given                                                                                                                                                                                                        |
|                     |           | 1–2 marks | a) Identifies a relevant risk which is specific to the task<br>b) Suggests measure(s) to manage the risk                                                                                                                        |
|                     |           | 3–4 marks | a) Identifies most of the relevant risks which are specific to the task<br>b) Method reflects how risks need to be managed                                                                                                      |
| <b>Overall plan</b> | 4         | 0 marks   | Gives no relevant method                                                                                                                                                                                                        |
|                     |           | 1–2 marks | a) Method is logically ordered to produce results<br>b) Chooses range of data/observations that would test the hypothesis                                                                                                       |
|                     |           | 3–4 marks | a) Method is logically ordered to produce results and includes an explanation of why it would test the hypothesis<br>b) Chooses range of data/observations that would test the hypothesis and explains why the range was chosen |
| <b>Total marks</b>  | <b>20</b> |           |                                                                                                                                                                                                                                 |

## Unit PCA: Physics controlled assessment

### Part B - Observations

| Element                               | Marks    |         | Criteria                                                                                                                                                            |
|---------------------------------------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary evidence and recording</b> | 4        | 0 marks | Collects no primary evidence                                                                                                                                        |
|                                       |          | 1 mark  | Records some data/observations that are appropriate for the topic                                                                                                   |
|                                       |          | 2 marks | Collects a suitable range of data/observations and records some appropriately (depends on the practical)                                                            |
|                                       |          | 3 marks | Collects a suitable range of data/observations and records all appropriately (depends on the practical)                                                             |
|                                       |          | 4 marks | Collects a suitable range of data/observations and records all appropriately (depends on the practical) and records further/repeat data                             |
| <b>Secondary evidence</b>             | 2        | 0 marks | Collects no secondary evidence                                                                                                                                      |
|                                       |          | 1 mark  | Collects and records secondary evidence relevant to the hypothesis in a way appropriate for the topic                                                               |
|                                       |          | 2 marks | Collects and records secondary evidence relevant to the hypothesis in a way appropriate for the topic. Comments on the quality of the sources of secondary evidence |
| <b>Total marks</b>                    | <b>6</b> |         |                                                                                                                                                                     |

# Part C - Conclusions

| Element                    | Marks |           | Criteria                                                                                                                                                                                                                                                                                                       |
|----------------------------|-------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Processing evidence</b> | 4     | 0 marks   | Evidence is not processed                                                                                                                                                                                                                                                                                      |
|                            |       | 1–2 marks | a) Attempts to process all collected evidence, using appropriate mathematical skills<br>b) Attempts to present the processed evidence in a way appropriate for the topic                                                                                                                                       |
|                            |       | 3–4 marks | a) Processes all collected evidence in a way that is appropriate to the task using appropriate mathematical skills<br>b) Presents processed evidence in a way that allows conclusions to be drawn                                                                                                              |
| <b>Quality of evidence</b> | 4     | 0 marks   | Makes no comments on the quality of the evidence                                                                                                                                                                                                                                                               |
|                            |       | 1–2 marks | a) Comments on the quality of the primary evidence, dealing with anomalies appropriately (if no anomalies in evidence candidates need to state this)<br>b) Comments on the quality of the secondary evidence, dealing with anomalies appropriately (if no anomalies in evidence candidates need to state this) |
|                            |       | 3–4 marks | a) Explains any adjustments to the evidence needed or decision not to exclude evidence<br>b) Takes account of anomalies in primary and secondary evidence when processing evidence (using all evidence if no anomalies)                                                                                        |

## Unit PCA: Physics controlled assessment

| Element                              | Marks |           | Criteria                                                                                                                                                                                                                                                          |
|--------------------------------------|-------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conclusions based on evidence</b> | 6     | 0 marks   | Makes no relevant conclusions                                                                                                                                                                                                                                     |
|                                      |       | 1–2 marks | a) Provides a conclusion based on all collected evidence, but does not link it to the hypothesis<br>b) Attempts to explain the conclusion using all collected evidence, including appropriate mathematical techniques                                             |
|                                      |       | 3–4 marks | a) Provides a conclusion which refers to the hypothesis based on all collected evidence<br>b) Explains the conclusion using the evidence, including appropriate mathematical relationships                                                                        |
|                                      |       | 5–6 marks | a) Provides a conclusion which refers to the hypothesis based on all collected evidence and relevant scientific ideas<br>b) Explains the conclusions using relevant scientific ideas and all collected evidence, including appropriate mathematical relationships |
| <b>Evaluation of conclusion</b>      | 4     | 0 marks   | Makes no relevant evaluation                                                                                                                                                                                                                                      |
|                                      |       | 1–2 marks | a) Evaluates conclusion based on all collected evidence<br>b) Suggests how all collected evidence can be improved to provide stronger support for the conclusion                                                                                                  |
|                                      |       | 3–4 marks | a) Evaluates conclusion based on all collected evidence and relevant scientific ideas<br>b) Suggests how all collected evidence can be improved and extended to provide stronger support for the conclusion                                                       |

## Unit PCA: Physics controlled assessment

| Element                     | Marks     |           | Criteria                                                                                                                                                                                                                                                                                            |
|-----------------------------|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Evaluation of method</b> | 6         | 0 marks   | Makes no relevant evaluation                                                                                                                                                                                                                                                                        |
|                             |           | 1–2 marks | a) Identifies a strength or weakness in the method<br>b) Suggests how to improve method and justifies comments made                                                                                                                                                                                 |
|                             |           | 3–4 marks | a) Describes strengths or weaknesses in the method and reasons for any anomalies<br>b) Suggests how to improve method and justifies comments made relating to the quality of the evidence collected (including reasons for anomalies)                                                               |
|                             |           | 5–6 marks | a) Describes strengths and weaknesses in the method and relates them to the hypothesis, and reasons for any anomalies<br>b) Suggests how to improve method, justifying comments made relating to the hypothesis and how better quality evidence could be produced (including reasons for anomalies) |
| <b>Total marks</b>          | <b>24</b> |           |                                                                                                                                                                                                                                                                                                     |

## B Assessment

### Assessment summary

Unit P1 is externally assessed by a one hour examination

Unit P2 is externally assessed by a one hour examination

Unit P3 is externally assessed by a one hour examination

Unit PCA is an internally assessed unit

### Summary of table of assessment

|                                                                                                                                                                                                                                                                                                                                                                                                   |                                |                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------|
| <b>Unit P1:</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>Universal physics</b>       | <b>Unit code: 5PH1F/5PH1H</b> |
| <ul style="list-style-type: none"><li>• This unit is assessed through a one hour, 60 mark, tiered written examination, containing six questions.</li><li>• The examination will contain a mixture of question styles, including objective questions, short answer questions and extended writing questions.</li><li>• Availability: June Series.</li><li>• First assessment: June 2014.</li></ul> |                                |                               |
| <b>Unit P2:</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>Physics for your future</b> | <b>Unit code: 5PH2F/5PH2H</b> |
| <ul style="list-style-type: none"><li>• This unit is assessed through a one hour, 60 mark, tiered written examination, containing six questions.</li><li>• The examination will contain a mixture of question styles, including objective questions, short answer questions and extended writing questions.</li><li>• Availability: June Series.</li><li>• First assessment: June 2014.</li></ul> |                                |                               |
| <b>Unit P3:</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>Applications of physics</b> | <b>Unit code: 5PH3F/5PH3H</b> |
| <ul style="list-style-type: none"><li>• This unit is assessed through a one hour, 60 mark, tiered written examination, containing six questions.</li><li>• The examination will contain a mixture of question styles, including objective questions, short answer questions and extended writing questions.</li><li>• Availability: June Series.</li><li>• First assessment: June 2014.</li></ul> |                                |                               |

**Unit PCA: Physics controlled assessment****Unit code: 5PH04**

- This unit is internally assessed under controlled conditions
- There will be two tasks available each year – one task from P2 and one task from P3
- Each task has a shelf life of one year
- The tasks will be available to teachers one year in advance
- Each task has three parts - Part A: Planning, Part B: Observations and Part C: Conclusions
- The total number of marks available for the three parts is 50
- Students must attempt all three parts of a task
- If they attempt both tasks, then the best marks from Part A, B and C should be submitted for the unit
- Availability: June Series
- First assessment: June 2014



## Assessment Objectives and weightings

|                                                                                                     | % in GCSE           |
|-----------------------------------------------------------------------------------------------------|---------------------|
| AO1: Recall, select and communicate their knowledge and understanding of physics                    | <b>33 – 39%</b>     |
| AO2: Apply skills, knowledge and understanding of science in practical and other contexts           | <b>34 – 40%</b>     |
| AO3: Analyse and evaluate evidence, make reasoned judgements and draw conclusions based on evidence | <b>25.5 – 28.5%</b> |
| <b>TOTAL</b>                                                                                        | <b>100%</b>         |

## Relationship of Assessment Objectives to units

| Unit                                    | Assessment Objective |                 |                     |                            |
|-----------------------------------------|----------------------|-----------------|---------------------|----------------------------|
|                                         | AO1                  | AO2             | AO3                 | Total for AO1, AO2 and AO3 |
| Unit P1: Universal physics              | 11 – 13%             | 7 – 9%          | 4.5 – 5.5%          | 25%                        |
| Unit P2: Physics for your future        | 11 – 13%             | 7 – 9%          | 4.5 – 5.5%          | 25%                        |
| Unit P3: Applications of physics        | 11 – 13%             | 7 – 9%          | 4.5 – 5.5%          | 25%                        |
| Unit PCA: Physics controlled assessment | 0%                   | 13%             | 12%                 | 25%                        |
| <b>Total for GCSE in Physics</b>        | <b>33 – 39%</b>      | <b>34 – 40%</b> | <b>25.5 – 28.5%</b> | <b>100%</b>                |

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## Entering your students for assessment

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### Student entry

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From summer 2014 onwards students will be required to sit all of their examinations and submit controlled assessment work for moderation at the end of the course. Students may complete the controlled assessment task(s) at any point during the course. As the controlled assessment task(s) changes each year, centres must ensure that they use the appropriate task for the year of GCSE entry.

Details of how to enter students for this qualification can be found in Edexcel's *UK Information Manual*; a copy is sent to all examinations officers. The information can also be found on Edexcel's website ([www.edexcel.com](http://www.edexcel.com)).

All externally assessed units will be assessed by tiered examinations. Students will need to be entered for a specific tier at the time of entry.

### Forbidden combinations and classification code

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Centres should be aware that students who enter for more than one GCSE qualification with the same classification code will have only one grade (the highest) counted for the purpose of the School and College Performance Tables.

Students should be advised that, if they take two qualifications with the same classification code, schools and colleges are very likely to take the view that they have achieved only one of the two GCSEs. The same view may be taken if students take two GCSE qualifications that have different classification codes but have significant overlap of content. Students who have any doubts about their subject combinations should check with the institution to which they wish to progress before embarking on their programmes.

## Access arrangements and special requirements

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Edexcel's policy on access arrangements and special considerations for GCE, GCSE and Entry Level is designed to ensure access to qualifications for all students (in compliance with the Equality Act 2010) without compromising the assessment of skills, knowledge, understanding or competence.

Please see the Edexcel website ([www.edexcel.com](http://www.edexcel.com)) for:

- the JCQ policy *Access Arrangements, Reasonable Adjustments and Special Considerations*
- the forms to submit for requests for access arrangements and special considerations
- dates for submission of the forms.

Requests for access arrangements and special considerations must be addressed to:

Special Requirements  
Edexcel  
One90 High Holborn  
London WC1V 7BH

## Equality Act 2010

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Please see the Edexcel website ([www.edexcel.com](http://www.edexcel.com)) for information with regard to the Equality Act 2010.

## Controlled assessment

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In controlled assessments, control levels are set for three linked processes: task setting, task taking and task marking. The control levels (high, medium or limited are dependent on the subject) are set for each process so that the overall level of control secures validity and reliability, provides good manageability for all involved and allows teachers to authenticate the student work confidently.

The summary of the controlled conditions for this qualification are shown below.

## Summary of conditions for controlled assessment

### Levels of control

Internal assessment under controlled conditions has levels of control for task setting, task taking and task marking. These must be adhered to when students are completing their controlled assessment tasks.

#### Summary of levels of control

| Area                  | Level of control |
|-----------------------|------------------|
| Part A – Planning     | Limited          |
| Part B – Observations | Limited          |
| Part C – Conclusions  | High             |

### Task setting

#### High level of control

A high level of control means that tasks will be set by Edexcel and centres will choose from a list of tasks, from the other units in this qualification.

#### When will the tasks be available?

They will be available on the Edexcel website for teachers to download a year ahead of their first assessment opportunity. Teachers can view all the task sheets available before deciding which task the students will complete.

#### When should the tasks be made available to students?

The task sheets for this controlled assessment are confidential and must not be shown to students before they start the tasks. Task sheets should not be shown to students until the start of the task planning stage of the controlled assessment.

#### Do all my students have to do the same task?

It is acceptable for all the students in a class to complete the same task. However, the same task does not have to be chosen for all students and they can work on a mixture of different tasks from P2 and P3.

The tasks will change every year, in accordance with the Ofqual regulations for GCSE Science. Teachers must take care when using these tasks to ensure that students are completing the correct task for a particular year. The front sheet of each task will show the dates for which it is valid.

### Task taking

#### a Research and data collection – limited level of control

Research and data collection, including practical work, will be carried out under limited control. This means that students may work collaboratively when collecting data from practical activities.

Students may carry out any secondary research whilst not being directly supervised by a teacher, for example in a library or at home. The secondary research can include extracts from books and websites.

**b Analysis, conclusions and evaluation of findings – high level of control**

The analysis, conclusions and evaluation will be produced by students under high levels of control. This means that this part of the write-up must be carried out individually by the students, under the supervision of a teacher.

The production of the final report will usually take place over several lessons, so the students' materials must be collected in at the end of the lesson and handed back at the beginning of the next one. Students' final reports must be produced individually.

## **Task marking**

### **Task marking – medium level of control**

A medium level of control means that the marking of the tasks will be carried out by teachers and moderated by Edexcel.

## **Internal standardisation**

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Teachers must show clearly how the marks have been awarded in relation to the assessment criteria. If more than one teacher in a centre is marking students' work, there must be a process of internal standardisation to ensure that there is consistent application of the assessment criteria.

## **Authentication**

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All students must sign an authentication statement. Statements relating to work not sampled should be held securely in your centre. Those that relate to sampled students must be attached to the work and sent to the moderator. In accordance with a revision to the current Code of Practice, any student unable to provide an authentication statement will receive zero credit for the component. Where credit has been awarded by a centre-assessor to sampled work without an accompanying authentication statement, the moderator will inform Edexcel and the mark will be adjusted to zero.

## **Further information**

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For more information on annotation, authentication, mark submission and moderation procedures, please refer to the *Edexcel GCSEs in Science, Additional Science, Biology, Chemistry and Physics: Instructions and administrative documentation for internally assessed units* document, which is available on the Edexcel website.



For up-to-date advice on teacher involvement, please refer to the Joint Council for Qualifications (JCQ) Instructions for conducting coursework/portfolio document on the JCQ website ([www.jcq.org.uk](http://www.jcq.org.uk)).

For up-to-date advice on malpractice and plagiarism, please refer to the Joint Council for Qualifications (JCQ) *Suspected Malpractice in Examinations: Policies and Procedures and Instructions for conducting coursework/portfolio* documents on the JCQ website ([www.jcq.org.uk](http://www.jcq.org.uk)).

## Assessing your students

The first assessment opportunity for all units of this qualification will take place in the June 2014 series and in each following June series for the lifetime of the specification.

### Your student assessment opportunities

#### GCSE in Physics

| Unit                                    | June 2014 | June 2015 | June 2016 | June 2017 |
|-----------------------------------------|-----------|-----------|-----------|-----------|
| Unit P1: Universal physics              | ✓         | ✓         | ✓         | ✓         |
| Unit P2: Physics for your future        | ✓         | ✓         | ✓         | ✓         |
| Unit P3: Applications of physics        | ✓         | ✓         | ✓         | ✓         |
| Unit PCA: Physics controlled assessment | ✓         | ✓         | ✓         | ✓         |

## Awarding and reporting

The grading, awarding and certification of this qualification will comply with the requirements of the current GCSE/GCE Code of Practice, which is published by the Office of Qualifications and Examinations Regulation (Ofqual). The GCSE qualification will be graded and certificated on an eight-grade scale from A\* to G. Individual unit results will be reported.

The first certification opportunity for the Edexcel GCSE in Physics will be in 2014.

Students whose level of achievement is below the minimum judged by Edexcel to be of sufficient standard to be recorded on a certificate will receive an unclassified U result.

## Unit results

### All units

The following table shows the uniform mark at each grade point for a unit that contributes 80 uniform marks towards the overall qualification.

| Unit grade      | A* | A  | B  | C  | D  | E  | F  | G  |
|-----------------|----|----|----|----|----|----|----|----|
| Foundation tier |    |    | 55 | 48 | 40 | 32 | 24 | 16 |
| Higher tier     | 72 | 64 | 56 | 48 | 40 | 36 |    |    |
| Un-tiered unit  | 72 | 64 | 56 | 48 | 40 | 32 | 24 | 16 |

Please note that a Foundation tier unit is restricted to the grades C–G. For this reason, the maximum uniform mark available is 1 uniform mark below grade B (55 uniform marks in this case).

A Higher tier unit is restricted to the grades A\*–D, with an allowed grade E; this allowed grade E being a 'half grade'. For this reason, the grade E uniform mark is set at 36 in this example.

Students who do not achieve the standard required for a grade G will receive a uniform mark in the range 0–15.

## Qualification results

The minimum uniform marks required for each grade:

GCSEs in Physics Cash-in code: 2PH01

| Qualification grade        | A   | B   | C   | D   | E   | F  | G  |
|----------------------------|-----|-----|-----|-----|-----|----|----|
| Maximum uniform mark = 320 | 256 | 224 | 192 | 160 | 128 | 96 | 64 |

Students who do not achieve the standard required for a grade G will receive a uniform mark in the range 0–63.

## Re-taking of qualifications

Students wishing to re-take a GCSE are required to re-take all the units in the qualification. Students will be permitted to carry forward the results from the controlled assessment unit(s) if they wish and only re-take the externally-assessed units.



## Language of assessment

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Assessment of this qualification will be available in English only. Assessment materials will be published in English only and all work submitted for examination and moderation must be produced in English.

## Quality of written communication

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Students will be assessed on their ability to:

- write legibly, with accurate use of spelling, grammar and punctuation, in order to make the meaning clear
- select and use a form and style of writing appropriate to purpose and to complex subject matter
- organise relevant information clearly and coherently, using specialist vocabulary when appropriate.

## Stretch and challenge

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Students can be stretched and challenged in units through the use of different assessment strategies, for example:

- using a variety of stems in questions – for example analyse, evaluate, discuss, compare, describe, explain
- ensuring connectivity between sections of questions
- a requirement for extended writing
- use of a wider range of question types to address different skills – for example open-ended questions, case studies, etc.

## Malpractice and plagiarism

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For up-to-date advice on malpractice and plagiarism, please refer to the Joint Council for Qualifications (JCQ) *Suspected Malpractice in Examinations: Policies and Procedures* document on the JCQ website ([www.jcq.org.uk](http://www.jcq.org.uk)).

## Student recruitment

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Edexcel's access policy concerning recruitment to our qualifications is that:

- they must be available to anyone who is capable of reaching the required standard
- they must be free from barriers that restrict access and progression
- equal opportunities exist for all students.



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## Progression

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- Students who successfully achieve this GCSE in Physics can progress onto a number of qualifications, such as the Level 3 GCE in Physics.
- Students could also progress onto an Edexcel BTEC Level 3 Applied Science qualification.
- Students could also progress into employment.

## Grade descriptions

### Physics

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | <p>Learners recall, select and communicate precise knowledge and detailed understanding of physics. They demonstrate a comprehensive understanding of the nature of physics, its laws, principles and applications and the relationship between physics and society. They understand the relationships between scientific advances, their ethical implications and the benefits and risks associated with them. They use scientific and technical knowledge, terminology and conventions appropriately and consistently showing a detailed understanding of scale in terms of time, size and space.</p> <p>They apply appropriate skills, including communication, mathematical, technical and observational skills, knowledge and understanding effectively in a wide range of practical and other contexts. They show a comprehensive understanding of the relationships between hypotheses, evidence, theories and explanations and make effective use of models, including mathematical models, to explain abstract ideas, phenomena, events and processes. They use a wide range of appropriate methods, sources of information and data consistently, applying relevant skills to address scientific questions, solve problems and test hypotheses.</p> <p>Learners analyse, interpret and critically evaluate a broad range of quantitative and qualitative data and information. They evaluate information systematically to develop arguments and explanations taking account of the limitations of the available evidence. They make reasoned judgments consistently and draw detailed, evidence-based conclusions.</p> |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>C</b></p> | <p>Learners recall, select and communicate secure knowledge and understanding of physics. They demonstrate understanding of the nature of physics, its laws, principles and applications and the relationship between physics and society. They understand that scientific advances may have ethical implications, benefits and risks. They use scientific and technical knowledge, terminology and conventions appropriately, showing understanding of scale in terms of time, size and space.</p> <p>They apply appropriate skills, including communication, mathematical, technical and observational skills, knowledge and understanding in a range of practical and other contexts. They show understanding of the relationships between hypotheses, evidence, theories and explanations and use models, including mathematical models, to describe abstract ideas, phenomena, events and processes. They use a range of appropriate methods, sources of information and data, applying their skills to address scientific questions, solve problems and test hypotheses.</p> <p>Learners analyse, interpret and evaluate a range of quantitative and qualitative data and information. They understand the limitations of evidence and use evidence and information to develop arguments with supporting explanations. They draw conclusions based on the available evidence.</p> |
| <p><b>F</b></p> | <p>Learners recall, select and communicate limited knowledge and understanding of physics. They show a limited understanding that scientific advances may have ethical implications, benefits and risks. They recognise simple interrelationships between physics and society. They use limited scientific and technical knowledge, terminology and conventions, showing some understanding of scale in terms of time, size and space.</p> <p>They apply skills, including limited communication, mathematical, technical and observational skills, knowledge and understanding in practical and some other contexts. They recognise and use hypotheses, evidence and explanations and can explain straightforward models of phenomena, events and processes. Using a limited range of skills and techniques, they answer scientific questions, solve straightforward problems and test ideas.</p> <p>Learners interpret and evaluate limited quantitative and qualitative data and information from a narrow range of sources. They can draw elementary conclusions having collected limited evidence.</p>                                                                                                                                                                                                                                                                             |

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## C Resources, support and training

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### Edexcel resources

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The resources from Edexcel provide you and your students with comprehensive support for our GCSE in Physics qualification. These materials have been developed by subject experts to ensure that you and your department have appropriate resources to deliver the specification.

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### Edexcel publications

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You can order further copies of the specification, Sample Assessment Materials (SAMs) and Teacher's Guide documents from:

Edexcel Publications  
Adamsway  
Mansfield  
Nottinghamshire NG18 4FN

Telephone: 01623 467 467  
Fax: 01623 450 481  
Email: [publication.orders@edexcel.com](mailto:publication.orders@edexcel.com)  
Website: [www.edexcel.com](http://www.edexcel.com)

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### Endorsed resources

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Edexcel also endorses some additional materials written to support this qualification. Any resources bearing the Edexcel logo have been through a quality assurance process to ensure complete and accurate support for the specification. For up-to-date information about endorsed resources, please visit [www.edexcel.com/endorsed](http://www.edexcel.com/endorsed).

Please note that while resources are checked at the time of publication, materials may be withdrawn from circulation and website locations may change.

## Edexcel support services

Edexcel has a wide range of support services to help you implement this qualification successfully.

**ResultsPlus** – ResultsPlus is an application launched by Edexcel to help subject teachers, senior management teams and students by providing detailed analysis of examination performance. Reports that compare performance between subjects, classes, your centre and similar centres can be generated in just a few clicks. Skills maps that show performance according to the specification topic being tested are available for some subjects. For further information about which subjects will be analysed through ResultsPlus, and for information on how to access and use the service, please visit [www.edexcel.com/resultsplus](http://www.edexcel.com/resultsplus).

**Ask the Expert** – To make it easier for you to raise a query with us online, we have merged our **Ask Edexcel** and **Ask the Expert** services.

There is now one easy-to-use web query form that will allow you to ask any question about the delivery or teaching of Edexcel qualifications. You'll get a personal response, from one of our administrative or teaching experts, sent to the email address you provide.

We'll also be doing lots of work to improve the quantity and quality of information in our FAQ database, so you'll be able find answers to many questions you might have by searching before you submit the question to us.

### Support for Students

Learning flourishes when students take an active interest in their education; when they have all the information they need to make the right decisions about their futures. With the help of feedback from students and their teachers, we've developed a website for students that will help them:

- Understand subject specifications
- Access past papers and mark schemes
- Find out how to get exams remarked
- Learn about other students' experiences at university, on their travels and entering the workplace

We're committed to regularly updating and improving our online services for students. The most valuable service we can provide is helping schools and colleges unlock the potential of their learners. [www.edexcel.com/students](http://www.edexcel.com/students)



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## Training

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A programme of professional development and training courses, covering various aspects of the specification and examination, will be arranged by Edexcel each year on a regional basis. Full details can be obtained from:

Training from Edexcel  
Edexcel  
One90 High Holborn  
London WC1V 7BH

Telephone: 0844 576 0027  
Email: [trainingbookings@edexcel.com](mailto:trainingbookings@edexcel.com)  
Website: [www.edexcel.com](http://www.edexcel.com)

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## D Appendices

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## Appendix 1: Codes

| Type of code                                  | Use of code                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Code number                                                                                 |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| National classification codes                 | Every qualification is assigned to a national classification code indicating the subject area to which it belongs. Centres should be aware that students who enter for more than one GCSE qualification with the same classification code will have only one grade (the highest) counted for the purpose of the school and college performance tables.                                                                                                       | 1210                                                                                        |
| National Qualifications Framework (NQF) codes | Each qualification title is allocated a National Qualifications Framework (NQF) code.<br><br>The National Qualifications Framework (NQF) code is known as a Qualification Number (QN). This is the code that features in the DfE Section 96 and on the LARA as being eligible for 16–18 and 19+ funding, and is to be used for all qualification funding purposes. The QN is the number that will appear on the student’s final certification documentation. | The QN for the qualification in this publication is:<br>GCSE in Physics – 600/0771/0        |
| Unit codes                                    | Each unit is assigned a unit code. This unit code is used as an entry code to indicate that a student wishes to take the assessment for that unit. Centres will need to use the entry codes only when entering students for their examination.                                                                                                                                                                                                               | Unit P1 – 5PH1F/5PH1H<br>Unit P2 – 5PH2F/5PH2H<br>Unit P3 – 5PH3F/5PH3H<br>Unit PCA – 5PH04 |
| Cash-in codes                                 | The cash-in code is used as an entry code to aggregate the student’s unit scores to obtain the overall grade for the qualification. Centres will need to use the entry codes only when entering students for their qualification.                                                                                                                                                                                                                            | GCSE in Physics – 2PH01                                                                     |
| Entry codes                                   | The entry codes are used to: <ul style="list-style-type: none"> <li>enter a student for the assessment of a unit</li> <li>aggregate the student’s unit scores to obtain the overall grade for the qualification.</li> </ul>                                                                                                                                                                                                                                  | Please refer to the <i>Edexcel UK Information Manual</i> , available on the Edexcel website |

## Appendix 2: How Science Works mapping

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